SIMATS ENGINEERING

CO5- ASSESSMENT TOOL 3 – Technical Presentation

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1709**

COURSE OUTCOME COVERED

CO5: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications.* (BTL 5)

Assessment Tool Weightage: 10%

Time Duration: 1 Hour

QUESTIONS: 5 **Total Marks:50**

Code of Conduct:

I ***Shaik. Abdul Razak (192524358)*** *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ***ShaiK. Abdul Razak***

1. System Analysis Presentation

Present a reverse engineering analysis of an AI-based system, explaining its architecture, components, and working process using diagrams.

2. Architecture Reconstruction Demo

Prepare a technical presentation to reconstruct the architecture of a given software/AI system, highlighting module interactions and data flow.

3. Algorithm & Logic Identification

Present how you identified algorithms and decision-making logic in a reverse-engineered system, including methodology and tools used.

4. Data Flow & Processing Pipeline

Explain through a presentation the data flow, processing pipeline, and communication mechanisms in a reverse-engineered distributed system.

5. Enhancement & Redesign Proposal

Deliver a technical presentation proposing improvements or redesign strategies for an existing system based on reverse engineering findings.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

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SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

Reverse Engineering of an AI-Based Fraud Detection System Using PySpark and RDD

CSA1709 – Artificial Intelligence

Abdul Razak Shaik (192524358)

Engineer to Excel

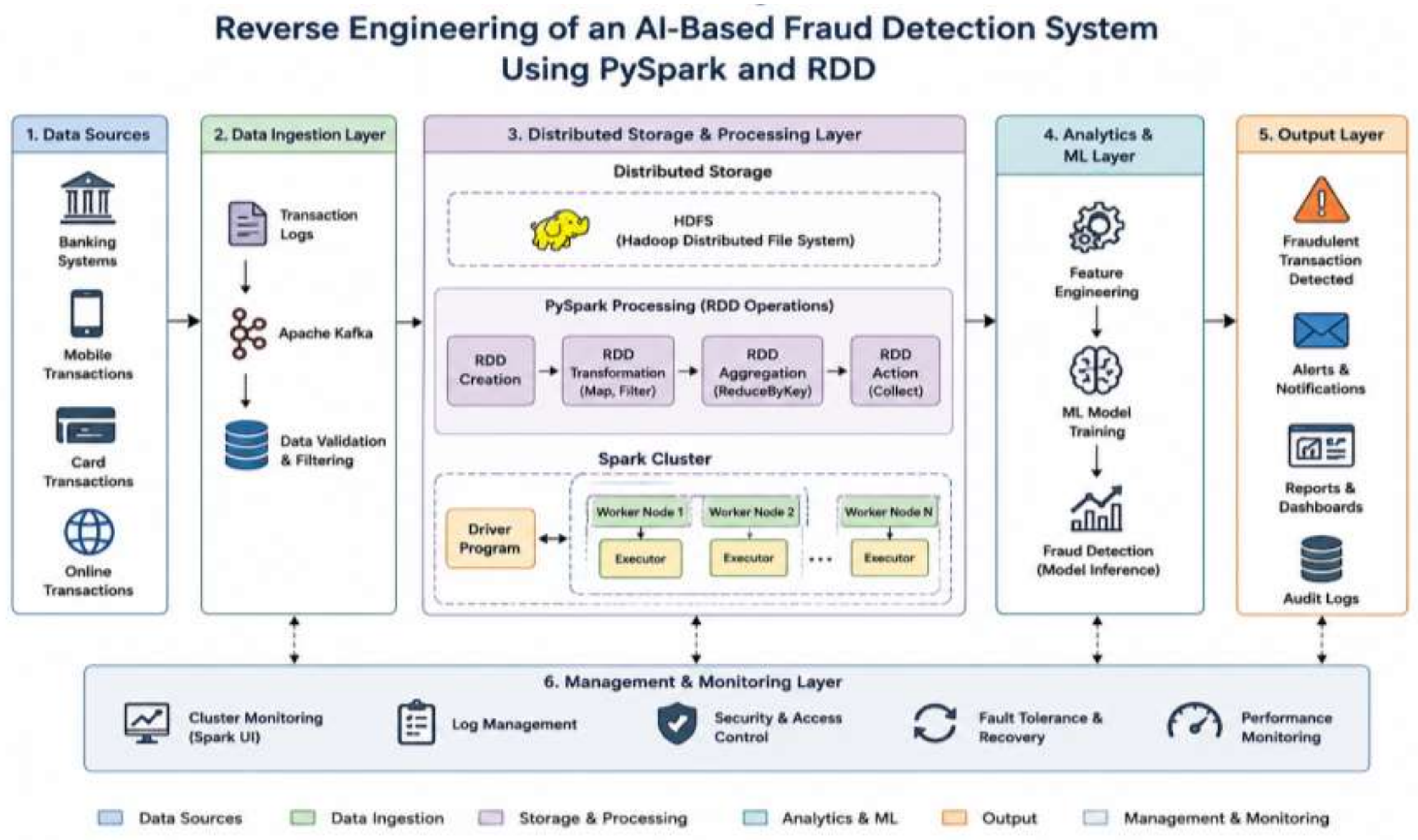
INTRODUCTION

- Banking systems process millions of transactions.
- Fraud detection identifies suspicious transactions.
- Distributed processing is required for large-scale data.
- PySpark provides scalable data processing.
- Reverse engineering helps understand the internal architecture

System Analysis

- Identify system inputs and outputs.
- Analyze major processing components.
- Study communication between modules.
- Identify data storage and processing mechanisms.
- Determine algorithms used for fraud detection.

System Architecture



Major Modules

- Transaction Data Acquisition
- Distributed Data Processing
- Feature Engineering
- Fraud Detection
- Alert and Reporting

Algorithm & Logic Identification

- RDD transformations
- RDD actions
- Map and Reduce operations
- Feature extraction
- Classification logic
- Fraud probability/threshold-based decision making

Communication Mechanisms

- Driver coordinates the application.
- Executors process data in parallel.
- RDDs distribute datasets across worker nodes.
- Spark communicates tasks between driver and executors.
- Distributed storage provides scalable data access.

Enhancement & Redesign

- Introduce real-time transaction processing.
- Use optimized Spark transformations.
- Add advanced ML models.
- Implement continuous fraud monitoring.
- Improve fault tolerance and scalability.
- Add real-time alert generation.

Conclusion

- Reverse engineering helps understand complex AI systems.
- PySpark enables distributed processing of large transaction datasets.
- RDD operations support parallel data processing.
- Identifying algorithms helps understand fraud detection decisions.
- The redesigned architecture improves scalability and real-time detection.

THANK YOU